

Charge Air Cooler (CAC) Cleaning



Material	
Item	Specification
Metal Brake Parts Cleaner PM-4-A or PM-4-B (US); CPM-4 (Canada)	—

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1. Remove the Charge Air Cooler (CAC) . For additional information, refer to Charge Air Cooler in this section.
2. **NOTICE: Do not use a high-pressure power washer to clean the Charge Air Cooler (CAC) or the CAC may be damaged.**

NOTE: *Drain all contaminates such as coolant, fuel and oil prior to cleaning the Charge Air Cooler (CAC) .*

NOTE: *Thoroughly clean the joint clamp areas as well as the turbocharger connections and the engine connection using metal brake parts cleaner.*

Lay the CAC flat with the inlet and outlet ports pointing up.
3. **NOTE:** *Plug or cap the CAC openings prior to agitating.*

Add an appropriate amount of commercially available detergent cleaner such as Simple Green® Pro HD, or equivalent, to the CAC . Follow the manufacturer's directions for cleaning. Fill the CAC to 40% of its volume with water.
4. Raise one end of the CAC and agitate it by hand for at least 5 minutes.
5. Raise the opposite end of the CAC and agitate it by hand for at least 5 minutes.
6. Drain the CAC .
7. Flush the CAC thoroughly with clean water.
8. Repeat Steps 1 through 6 until no contaminates are found in the flush water.
9. Allow the CAC to air dry.
10. **NOTE:** *The following leak test steps must be performed prior to installing the CAC .*

NOTE: *Use a commercially available kit, such as the SPX/OTC Charged Air Cooler Tester No. 5039, or equivalent.*

Install the commercially available CAC cooler tester on the CAC following the manufacturer's installation instructions.
 - Tighten the clamps to 5 Nm (44 lb-in).
11. **WARNING: Never exceed the specified pressure. Excessive pressure may cause the test adapter to blow off or may damage the charge air cooler (CAC). Failure to follow this instruction may result in serious personal injury.**

Slowly apply air pressure to 150 kPa (22 psi).

12. Let the CAC stand for a few minutes and note any loss in pressure.

- Release the air pressure.

13. Repeat Steps 11 and 12 as many times as necessary to verify the readings. The reading is considered verified when 3 consecutive tests show approximately the same pressure drop.

- If the pressure loss exceeds 20 kPa (3 psi) per minute, install a new CAC. For additional information, refer to Charge Air Cooler in this section.